

1. Language

We'll start by giving a formal account of the simplest practical language – that of *first-order* arithmetic expressions and function calls, like $x + y$ and $a + (5 * b) + \sin(3)$.

Rather than complicate our language by introducing binary operators (like $x + y$), unary operators (like $-x$), and n -ary function applications (like $f(x, y, z)$), we can instead normalize everything to *S-expressions* – or *sexprs* – of the form $(f e_1 \dots e_n)$. We can then uniformly write:

- binary operations like $x + y$ as binary sexprs $(+ x y)$,
- unary operations $-x$ as unary sexprs $(- x)$,
- function applications $f(x, y, z)$ as n -ary sexprs $(f x y z)$.

In particular, the *operator* f always comes first, and there is no order-of-operations – all bracketing is explicit. Restricting ourselves to *first-order* sexprs means that we always require operators f to be atomic symbols drawn from a fixed set O – in particular, we don't support *partial application*, like $((\text{add } 2) 3)$, since that would require $f = (\text{add } 2)$ to be a compound expression.

Formally, we define our *syntax* as follows:

Definition 1.1 (First-order S-expressions): fixing

- a set of *variables* $x \in N$
- a set of *operations* $f \in O$
- a set of *constants* $c \in V$

we define the set of *first-order S-expressions* $e \in \text{STm}_N(O)$ to be given by the grammar

$$e ::= x \mid c \mid (f e^*)$$

TODO 1: Open-vs-closed programs – operational semantics of closed programs first

TODO 2: Operational semantics of S-expressions – partiality via division

TODO 3: Observational equivalence of S-expressions

TODO 4: Well-typed vs. ill-typed S-expressions

Given a set of *types* $A \in \mathcal{T}$, we'll define a *stack-typing relation* on O to be any relation

$$f : [A_1, \dots, A_m] \rightarrow [B_1, \dots, B_n]$$

where

- $A_1, \dots, A_m \in \mathcal{T}$ are types – m is called the *input arity* or *arity* of f .
- $B_1, \dots, B_n \in \mathcal{T}$ are types – n is called the *output arity* of f .

For now, we'll only consider operations with an output arity of 1.

We define a (*typing*) *context* $\Gamma \in \text{Ctx}_N(\mathcal{T})$ to be given by a list of *bindings*

$$x_1 : A_1, \dots, x_n : A_n$$

We can define a simple type system for S-expressions by giving a typing relation $\Gamma \vdash e : A$ that says “under the assumptions of Γ , the S-expression e has type A ”. The rules are:

TODO 5: Typing rules for constants – as distinct from nullary ops

$$\text{var-head} \frac{}{\Gamma, x : A \vdash x : A} \quad \text{var-tail} \frac{\Gamma \vdash y : B, x \neq y}{\Gamma, x : A \vdash y : B}$$

$$\text{app} \frac{f : [A_1, \dots, A_n] \rightarrow [B] \quad \forall i. \Gamma \vdash a_i : A_i}{\Gamma \vdash (f \ a_1 \ \dots \ a_n) : B}$$

TODO 6: *Soundness* of typing: *preservation of typing*

TODO 7: *Completeness* of typing: in the presence of *simple types*

TODO 8: Church vs. Curry – we will only consider *well-typed* terms

TODO 9: Doesn't lose generality – untyping works in this framework due to multi-arity

TODO 10: *Substitution* and *observational equivalence* – typing rules for substitution – two open programs are equivalent iff equivalent under every substitution – this is usually where *metavariables* come in

TODO 11: *Purity* – required for substitution to preserve equivalence – typing rules for purity

TODO 12: Denotational semantics of S-expressions: partial functions

TODO 13: Denotational semantics of S-expressions: *soundness* and *completeness*

TODO 14: Equational theory of S-expressions: *purity* – as otherwise you can't *introduce* divisions

TODO 15: Equational theory of S-expressions: *soundness* and *completeness*

2. State

TODO 16: Operational semantics of S-expressions – *state*

TODO 17: *Observational equivalence* in the presence of state

TODO 18: Denotational semantics of S-expressions – the partial state monad

TODO 19: Denotational semantics of S-expressions: *soundness* and *completeness*

TODO 20: Equational theory of S-expressions: *purity*

TODO 21: Equational theory of S-expressions: *soundness* and *completeness*